

Intelligent Cognitive AlarmPlatform

Ratnesh Kumar

Table of Contents

1. Project Description	1
1. Dataset Used	1
3. Environment Setup	2
4. Data Exploration	3
5. Data Preprocessing	3
6. Proposed System	4
7. Training Pipeline	5-11
8. Conclusion	12

1. Project Description

The Intelligent Cognitive Alarm Platform is an AI-powered alarm management system designed to help users develop healthy wake-up habits and improve productivity. Unlike traditional alarm applications, users must complete cognitive challenges such as math problems, memory games, logic puzzles, or riddles before dismissing an alarm. The platform analyzes user behavior, sleep patterns, and alarm usage to encourage consistent wake-up routines. The backend is developed using **FastAPI** with **PostgreSQL**, while the frontend can be built using **React.js** or React Native.

2. Technology Stack

The project uses **FastAPI** as the backend framework due to its high performance and automatic API documentation. **PostgreSQL** is used as the primary relational database for storing user and alarm information. **SQLAlchemy** serves as the **ORM**, while Alembic manages database migrations. **JWT** is used for secure authentication, and **Passlib (bcrypt)** ensures passwords are securely hashed. Docker is used for containerization, and Git/GitHub is used for version control.

3. Environment Setup

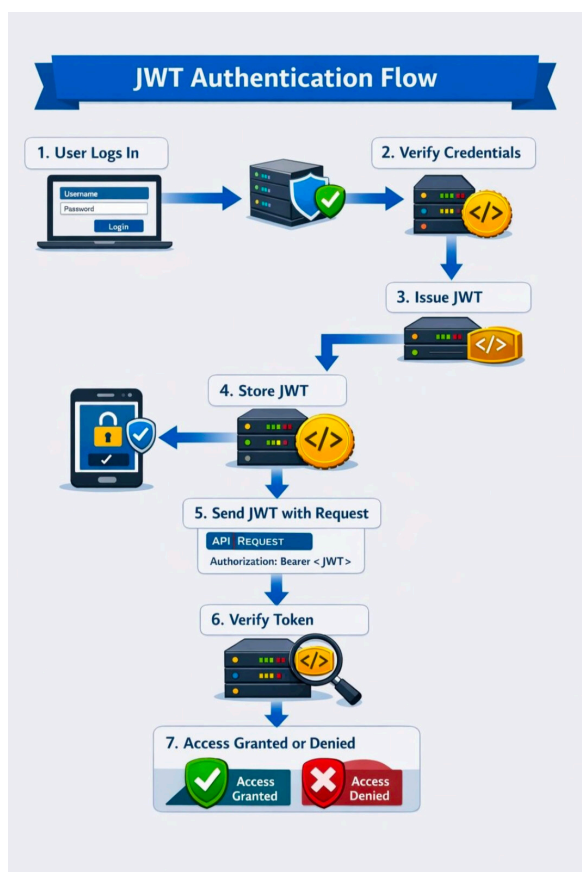
The development environment includes Python 3.12, **FastAPI**, **PostgreSQL**, **SQLAlchemy**, Alembic, **Uvicorn**, and Visual Studio Code. Required dependencies are installed using pip, environment variables are stored in a **.env** file, and the backend server is executed using Uvicorn. PostgreSQL is configured as the main database, and Swagger UI is available for testing **APIs**.

4. Database Design

The database consists of tables such as Users and Alarms. The Users table stores user details including profile information, authentication data, and roles. The Alarms table stores alarm information such as title, alarm time, repeat schedule, alarm type, and user association. SQLAlchemy models define relationships between tables, while Alembic manages schema updates.

5. Authentication Module

The authentication module provides secure user access using JWT authentication. Users can register and log in with their email and password. Passwords are encrypted using bcrypt before storage. JWT access tokens are generated after successful login and are required to access protected resources. The system also supports role-based access control for User, Admin, and Wellness Coach roles.



6. User Profile Management

The User Profile module allows users to create, view, update, and manage their personal information. Profile details include preferred wake-up time, sleep duration, time zone, productivity goals, difficulty preference, and habit preferences. This information helps personalize future alarm behavior and recommendations.

- User Registration
- User Login
- User Profile Creation
- Profile Information Management
- Personal Details Management
- Preferred Wake-up Time Settings
- Sleep Duration Management
- Time Zone Configuration
- Productivity Goal Management
- Difficulty Preference Settings
- Habit Preference Management
- Profile Update
- Profile Retrieval
- Profile Deletion
- User Role Management
- Profile Validation
- Database Design
- Security and Privacy



7. Alarm Scheduling

The Alarm Scheduling module allows users to create and manage multiple alarms. It supports Daily, Weekday, Weekend, One-Time, and Smart Adaptive alarms. Users can update, delete, activate, or deactivate alarms according to their schedule. Each alarm is linked to a specific user and stores information such as alarm time, repeat settings, and status.

Alarm Creation

Alarm Types

Alarm Configuration

Recurring Alarm Scheduling

One-Time Alarm Scheduling

Multiple Alarm Management

Alarm Activation and Deactivation

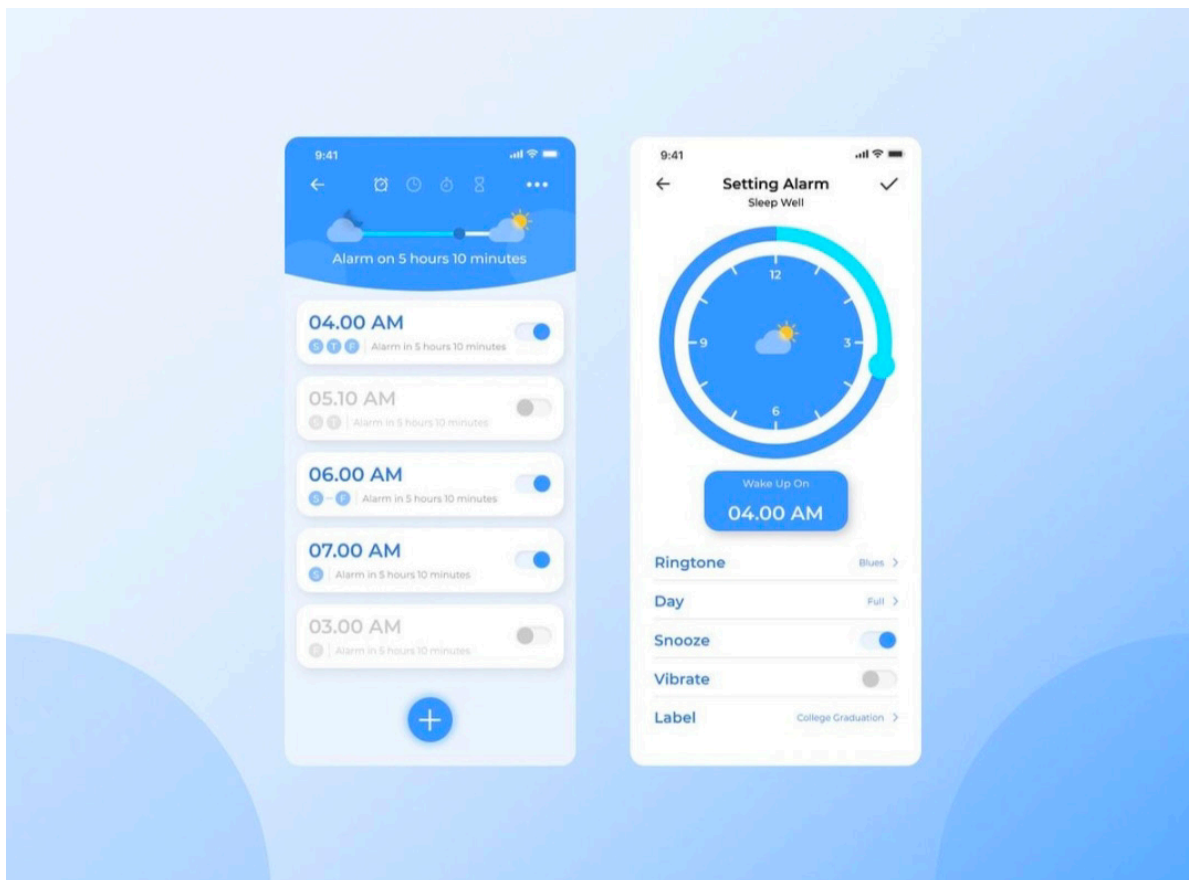
Alarm Validation

Time Zone Management

User-Alarm Relationship

Alarm Database Design

Alarm Scheduling Workflow



8. Conclusion

The Intelligent Cognitive Alarm Platform establishes a secure and scalable backend for intelligent alarm management. By integrating authentication, user profiles, and alarm scheduling, the system provides a solid base for future modules such as cognitive challenges, adaptive difficulty, behavioral analytics, and AI-powered recommendations. This architecture ensures maintainability, security, and scalability for real-world deployment.

Web Application Architecture

